

GlucoTwin — A Digital Twin for Proactive Glucose-Spike Risk Management

Healthcare Problem

- Diabetes monitoring is often retrospective.
- GlucoTwin combines historical patient context with current physiological and behavioral signals.
- Target outcome: probability of a high-glucose event in the next 2 hours.

Healthcare Use Case

- Type 2 diabetes risk management.
- Historical: age, BMI, HbA1c, diabetes duration, BP, family history, adherence and baseline glucose.
- Dynamic: glucose, slope, trajectory, heart rate, HRV, activity and sleep.
- Output: 2-hour probability + LOW / MODERATE / HIGH tier for clinician review.

What Makes It a Digital Twin?

- Static patient profile + dynamic observations + feature fusion + repeated state updates + time-bound prediction.
- The patient's represented state can change as new observations arrive.

Technical Architecture

- Historical/EHR + dynamic stream → feature engineering → digital-twin state vector → ML prediction → risk output → clinician dashboard → human review.
- Patient-level splitting reduces leakage across longitudinal observations.

Technical Stack

- Python 3.11+, Pandas, NumPy, scikit-learn, Streamlit, Plotly and Joblib.
- Reproducible synthetic data generator and public GitHub codebase.

AI / ML Methodology

- Logistic Regression baseline; Random Forest primary comparison.
- Temporal glucose statistics plus activity, HR, HRV and static profile features.
- Metrics: ROC-AUC, PR-AUC, precision, recall and F1.
- Synthetic-data results are development evidence, not clinical validation.

Live Prototype

- Run: `streamlit run app.py`
- Dashboard shows 2-hour risk, current glucose, HbA1c, trajectory, historical profile and dynamic signals.
- Human review remains mandatory.

Outcomes & Evidence

- End-to-end PoC: generation → feature fusion → validation → model selection → prediction → dashboard.
- Static and dynamic data are fused into a patient-specific state.
- Risk is localized to the next 2 hours.
- Next step: validate on de-identified real longitudinal data.

Safety & Responsible Deployment

- No diagnosis, prescribing or autonomous treatment.
- Before deployment: external validation, calibration, subgroup analysis, prospective evaluation, privacy/security controls and clinical governance.

Scale-Up Path

- Connect validated feeds → calibrate thresholds → personalize baselines → monitor drift/fairness/calibration → prospective outcome evaluation.

Closing

GlucoTwin turns fragmented health signals into a continuously updated risk state for proactive diabetes risk management.